

Project Confirmation & Initial Report

1. Basic Information

Project Title: Lifely AI – Intelligent Care Management & Personal Assistant for People with Physical Disabilities.

Full Name: Shivanshu Gupta

Student ID: 22f3002985

Term: May 2026

2. Problem Statement

People with physical disabilities often face significant challenges in managing their daily lives independently. They must regularly keep track of medications, doctor appointments, rehabilitation exercises, daily routines, personal goals, and health conditions. At the same time, caregivers and family members need a convenient way to monitor the individual's progress and provide assistance when required. Existing solutions usually focus on only one aspect of care, such as reminders, appointment scheduling, or note-taking, forcing users to switch between multiple applications. This fragmentation makes care management difficult and reduces overall efficiency.

Lifely AI is designed to solve this problem by providing a unified care management platform specifically tailored for individuals with physical disabilities. The platform combines daily health tracking, care plan management, goal tracking, medication reminders, appointment scheduling, secure document storage, caregiver collaboration, and an AI-powered assistant within a single application.

The primary users of the application are people with physical disabilities who require structured daily support. Secondary users include caregivers, parents, or family members responsible for monitoring and assisting these individuals. Administrators manage the platform, verify users, oversee caregiver assignments, and monitor overall system health.

The application addresses an important gap in current digital healthcare solutions by integrating multiple care management functions into one centralized platform. Unlike general reminder applications or hospital portals, Lifely AI focuses on long-term personal care, independence, and collaboration between users and caregivers. The AI assistant further enhances the experience by answering questions, summarizing health information, providing motivation, and offering personalized suggestions based on user data.

The overall objective of Lively AI is to improve independence, simplify communication between users and caregivers, reduce missed medications or appointments, and help users maintain consistent progress toward their personal health and rehabilitation goals.

3. System Overview

Lively AI is a full-stack web application that enables users with physical disabilities to manage their daily care while allowing caregivers to monitor progress and provide assistance through a secure role-based system.

The application consists of the following major components:

Authentication Module

- User Registration
- Login
- JWT Authentication
- Password Hashing
- Role-Based Access Control

Care Management Module

- Care Plans
- Daily Check-ins
- Goal Tracking
- Medication Management
- Appointment Scheduling

Caregiver Module

- Assigned User Monitoring
- Care Notes
- Health Progress Tracking
- Reminder Management

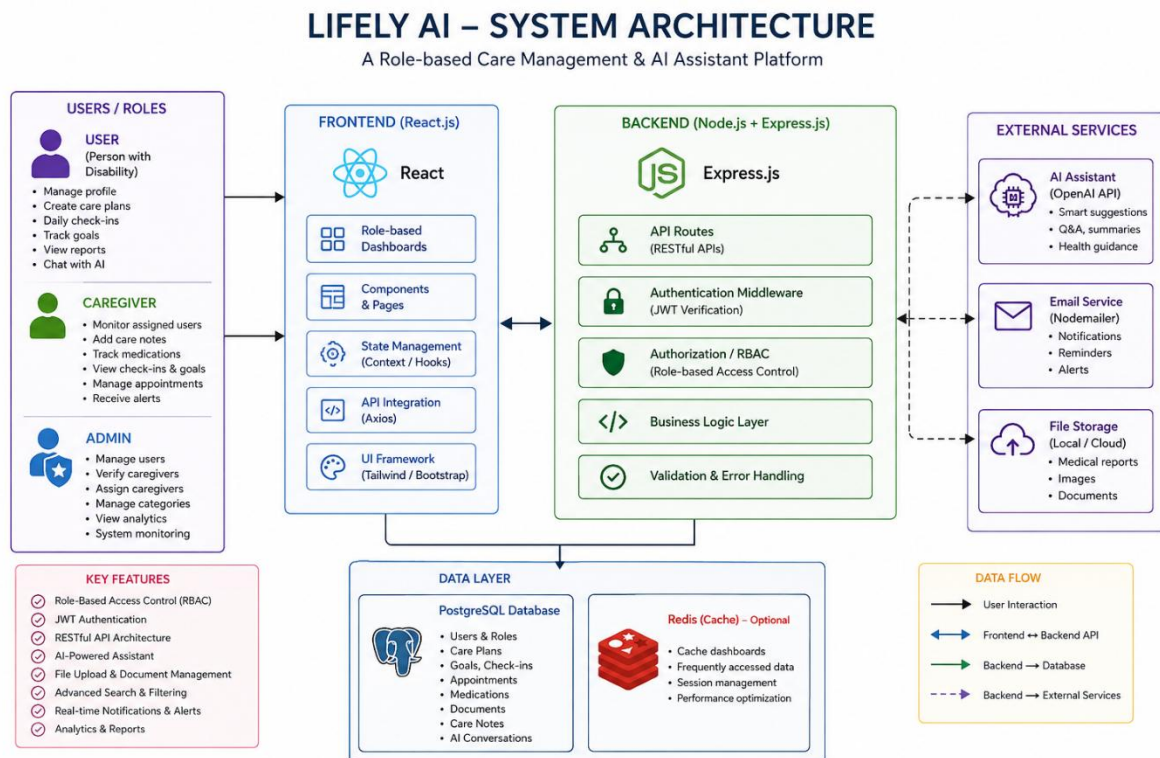
AI Assistant Module

- Ask health-related questions
- Daily summaries
- Goal suggestions
- Personalized recommendations

Administration Module

- User Management
- Caregiver Assignment
- Analytics Dashboard
- System Monitoring

High-Level Architecture



4. User Roles and Responsibilities

4.1 Admin (Superuser)

The Admin is responsible for managing the entire Lively AI platform and ensuring that the system operates securely and efficiently. The admin oversees user accounts, manages caregiver assignments, monitors platform activities, and enforces role-based access control. The admin dashboard provides an overview of system statistics and analytics, enabling effective monitoring of the application's performance.

Responsibilities:

- Manage user and caregiver accounts.
- Assign caregivers to users.
- View platform analytics and system reports.
- Manage categories and application settings.
- Monitor user activities and resolve issues.
- Remove or suspend unauthorized or inactive accounts.

Admin Dashboard Includes:

- Total registered users and caregivers.
- Active Care Plans.
- Daily Check-in statistics.

- Goal completion analytics.
- Appointed and activity summaries.

4.2 Role Type A – User (Person with Disability)

The **User** is the primary user of the platform who manages their daily care and health activities. Users can create personalized care plans, record health updates, schedule appointments, upload medical documents, and interact with the AI assistant for guidance and motivation. The dashboard is personalized to help users easily track their daily routine and progress.

Responsibilities:

- Create and manage Care Plans.
- Record daily health check-ins.
- Set and track personal goals.
- Manage medications and appointments.
- Upload medical reports and documents.
- Chat with the AI Assistant for personalized support.

User Dashboard Includes:

- Today's Care Plan.
- Upcoming medications and appointments.
- Goal progress tracker.
- Daily mood and health summary.
- AI recommendations.

4.3 Role Type B – Caregiver

The **Caregiver** supports users by monitoring their health progress and assisting with daily care activities. Caregivers can only access the profiles of users assigned to them, ensuring privacy and security. They help users stay on track by reviewing health updates, adding care notes, and receiving alerts for missed activities.

Responsibilities:

- Monitor assigned users' health progress.
- View Care Plans, goals, and daily check-ins.
- Add care notes and recommendations.

- Schedule appointments for assigned users.
- Receive alerts for missed medications or check-ins.
- Upload medical documents when required.

Caregiver Dashboard Includes:

- List of assigned users.
- Recent health updates.
- Missed check-in and medication alerts.
- Upcoming appointments.
- Care Plan progress and activity summaries.

5. User Roles and Responsibilities

Core Entity Name

Care Plan

Description

The Care Plan is the central entity of the system. It represents a personalized plan that helps users manage their daily activities, rehabilitation, medication schedules, or health goals.

6. Feature Implementation Status

Feature	Status
Multi-role System	Planned
Core Business Entity	Planned
Authentication & Authorization	Implemented
Role-based Dashboards	Planned
Care Plan Management	Planned
Daily Check-ins	Planned
Goal Tracking	Planned

Medication Management	Planned
Appointment Scheduling	Planned
File Upload	Planned
Advanced Search & Filtering	Planned
AI Assistant	Planned
Analytics Dashboard	Planned
Caregiver Monitoring	Planned
Notifications	Planned

7. Tech Stack Used and Purpose

Technology	Purpose
Express.js	Build RESTful APIs and implement backend business logic.
PostgreSQL	Store users, care plans, medications, appointments, and related data in a relational database.
TypeORM	Define entities, manage relationships, perform database migrations, and simplify database operations.
React.js	Build a responsive frontend with role-specific dashboards and reusable components.
JWT Authentication	Secure user authentication and API authorization using role-based access control.
Redis	Cache frequently accessed dashboard and Care Plan data to improve performance.
bcrypt	Securely hash user passwords before storing them.
Multer	Upload medical reports, prescriptions, and Care Plan images.
Axios	Enable communication between the React frontend and Express backend.
Chart.js / Recharts	Display analytics such as mood trends, goal completion, and health progress.